ST MARY'S DSG, KLOOF

GRADE: 10 **DATE**: 6 February 2019

MATHEMATICS – No Systems, Patterns, Algebra

TIME: 40 min **TOTAL**: 53

Question 1

a) For which values of x where $\boldsymbol{x \in \{-2\,;0\,;2\,;4\}}$ is $\sqrt{\frac{9}{2-x}}$ (4)

1) A irrational number? _____

2) A non real number? _____

3) A rational number ? _____

4) Undefined _____

b) Represent the following on a number line:

1) $\{x:\ -4 \leq x \leq 1\ ;\ x \in N_0\}$ (2)



2) $[-10\,;20)$ (2)



[8]

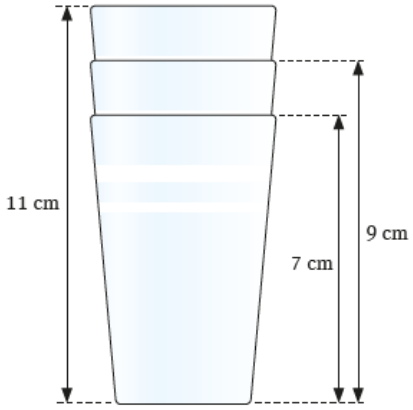
Question 2

a) Fill in the missing spaces in the following number patterns:

1) -1 ; 2 ; 5 ; _____ ; 11 ;_____ (2)

2) 2 ; 3 ; 5 ; 8 ; 12 ; _____ ; _____ (2)

b) You are stacking polystyrene cups, which fit into each other as in the diagram below. The 1st cup is 7 cm high. Two cups stacked together are 9 cm , and three cups stacked together are 11 cm high.



1) How high would 4 cups be ? (1)

_____

2) Find the genera formula for the height of a stack of n cups. (2)

_____

_____

3) How many cups will stack on a shelf if the height between the shelves is 21 cm ? (2)

_____

_____

4) A box is 30 cm high and can only fit a single stack of cups. You need 100 cups for a party, how many boxes would you need to buy? (3)

_____

_____

c) **4 ; 7 ; 14 ; 25……**

1) Determine the next 2 terms (2)

_____

_____

2) Determine the general term in the form of $T_n = an^2 + bn + c$ (4)

d) Consider the following number Pattern: $\mathbf{0,\dot{1}}$; $\mathbf{0,\dot{2}}$; $\mathbf{0,\dot{3}}$ ... ...

1) Determine the next two terms_____ (2)

2) Write down T_3 in the form $\frac{a}{b}$ (3)

3) Write down 2 rational numbers between the 2nd and 3rd term. (2)

4) Determine the largest number in the sequence where $T_n < 1$. (1)

[26]

Question 3

a) Multiply and simplify

1) $5\left(\frac{1}{5}x - 1\right)\left(5x - \frac{1}{5}\right)$ (3)

2) $-(2x - 3)(4x^2 + 6x + 9)$ (3)

3) $(3^x + 1)^2$ (2)

b) Factorise fully

1) $3a^2 - 12\,b^2$ (2)

2) $(a - 2b)^3 - 4\,(a - 2\,b)$ (3)

[13]

Question 4

a) If **0 ; 7 ; 26 ; 63 ; 124……..**
Determine the general term of the number pattern. (2)

b) A magic substance doubles it volume every minute. A small quantity is placed in a bottle and at 17h13 is exactly full. At what time is the container a quarter full (2)

c) Find the product of : (2)

$$\left(1-\frac{1}{3}\right)\left(1-\frac{1}{4}\right)\left(1-\frac{1}{5}\right)\left(1-\frac{1}{6}\right)\left(1-\frac{1}{7}\right)\ldots\ldots\ldots\ldots\ldots\ldots\left(1-\frac{1}{74}\right)\left(1-\frac{1}{75}\right)$$

[6]